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\*\*\*Original Research\*\*\*

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## **Mathematical Modeling and Optimal Cost-Effective Control of Mesquite Invasion in Coupled Land – Water Systems**

Authors

Joel Simon

Department of Mathematics and ICT

The Open University of Tanzania

ORCID: <https://orcid.org/0000-0002-5817-22X>

&

Mary Swai

Department of ICT and Mathematics

Open University of Tanzania

ORCID: <https://orcid.org/0000-0002-5817-29X>

&

Linus Kisoma

Department of Mathematics and Statistics

Sokoine University of Agriculture

ORCID: <https://orcid.org/0000-0002-5817-29X>



## Abstract

Mesquite (*Prosopis juliflora*) invasion has become a major ecological and socio-economic challenge in arid and semi-arid ecosystems due to its rapid spread through interconnected land, water, and animal-mediated dispersal pathways. Unlike previous studies that primarily considered isolated invasion mechanisms or single control interventions, this study develops an integrated mathematical model combining ecological transmission dynamics, sensitivity analysis, optimal control, and cost-effectiveness evaluation. Model parameterization was based on field observations conducted in Kahe Ward and published ecological studies. Parameters lacking direct empirical measurements, namely:  $\mu_1, \mu_2, \mu_F, \eta_1, \alpha_1, \beta_1, \beta_2$  and  $\rho_1$  were estimated by calibrating the model to observed field data in MATLAB using the adaptive Runge–Kutta (RK45) numerical solver, whereas the remaining parameters were adopted from previous studies. The model was analyzed through positivity, boundedness, equilibrium analysis, and invasion threshold derivation using the Next Generation Matrix approach. Sensitivity analysis identified the ecological parameters with the greatest influence on invasion persistence. Optimal control strategies were derived using Pontryagin's Maximum Principle and solved numerically using the fourth-order Runge–Kutta method. Simulation results showed that mesquite invasion increased steadily without intervention, whereas integrated adaptive control substantially reduced invasion across all transmission pathways. Cost-effectiveness analysis further demonstrated that although the combined control strategy incurred the highest implementation cost of USD 347.54 hectares per year, it achieved the lowest ecological burden of 2.1048 hectares per year, confirming integrated adaptive management as the most sustainable long-term strategy for mesquite invasion control.

**Keywords:** *Optimal control, Cost-effectiveness analysis, Integrated management, Mathematical modeling, Invasive species control*

## 1.0 Introduction

Biological invasions are among the most significant global threats to biodiversity, ecosystem functioning, and sustainable land use. Invasive plant species disrupt ecological processes, displace native vegetation, reduce ecosystem productivity, and impose substantial economic burdens on affected communities. These impacts are particularly pronounced in arid and semi-arid ecosystems, where harsh environmental conditions favor the establishment, persistence, and



rapid spread of drought-tolerant invasive species. Consequently, invasive alien plants have become a major concern for environmental conservation, agricultural productivity, and sustainable natural resource management worldwide.

Among the most aggressive invasive woody plants are mesquite (*Prosopis* spp.), particularly *Prosopis juliflora* and *Prosopis pallida*, which have become major ecological and socio-economic challenges across Africa, Asia, and Australia (Shackleton et al., 2022; Holden et al., 2024). These species were deliberately introduced into many countries for land rehabilitation, erosion control, fuelwood production, and livestock fodder. However, their exceptional adaptability, deep rooting systems, prolific seed production, and tolerance to prolonged drought have enabled them to spread far beyond their intended areas of introduction (Holden et al., 2024). As a result, mesquite forms dense infestations that suppress native vegetation, reduce grazing capacity, alter hydrological processes, degrade ecosystem services, and accelerate biodiversity loss (Hussain et al., 2023). These ecological impacts are accompanied by substantial socio-economic consequences, including reduced rangeland productivity, increased management costs, and declining livelihoods for pastoral and agro-pastoral communities that depend on healthy ecosystems.

The successful invasion of mesquite is facilitated by multiple interacting dispersal pathways operating across terrestrial and aquatic environments. Seeds are dispersed over long distances by livestock and wildlife after passing through their digestive systems, while seasonal floods, rivers, and surface runoff transport seeds through water systems, creating interconnected land-water invasion pathways that promote colonization and persistence (Epanchin-Niell & Wilen, 2022; Perrings et al., 2023). Furthermore, anthropogenic disturbances such as overgrazing, land degradation, deforestation, and land-use change increase habitat susceptibility and accelerate invasion by creating favorable conditions for seed establishment and population expansion. The interaction between ecological processes and human activities makes mesquite invasion highly complex and difficult to predict using conventional ecological approaches alone.

To mitigate these impacts, a range of management strategies has been implemented, including mechanical removal, chemical treatment, biological control, and integrated management approaches. Although these interventions can reduce invasion intensity, their effectiveness is often constrained by high implementation costs, reinvasion following treatment, environmental concerns, and variability in performance across ecosystems. Mechanical and chemical methods



generally provide only temporary suppression, whereas biological control may require long periods before producing measurable ecological benefits and often exhibits inconsistent effectiveness under different environmental conditions (Khumalo & Mwangi, 2023; Muturi & Muge, 2023). These limitations highlight the need for integrated management strategies that simultaneously consider ecological effectiveness, operational feasibility, and economic sustainability.

Mathematical modelling has become an indispensable tool for investigating ecological systems and supporting evidence-based management of invasive species. Deterministic differential equation models provide a rigorous quantitative framework for describing invasion dynamics, identifying dominant dispersal pathways, evaluating invasion thresholds, and predicting long-term system behaviour under different ecological conditions. Moreover, sensitivity analysis enables the identification of the parameters that most strongly influence invasion persistence, while optimal control theory provides systematic approaches for determining intervention strategies that minimize invasion while reducing implementation costs (Lin & Chen, 2022; Hattaf & Yousfi, 2023). Coupled with numerical simulation and cost-effectiveness analysis, these mathematical techniques offer valuable insights for designing efficient, sustainable, and economically viable management policies.

Despite substantial progress in invasive species modelling, most existing mathematical models of mesquite invasion consider only single dispersal mechanisms or evaluate individual control interventions independently. Relatively few studies integrate terrestrial and aquatic dispersal pathways, animal-mediated seed transmission, optimal control, sensitivity analysis, and economic evaluation within a unified mathematical framework. This limitation restricts the ability of current models to capture the complex ecological interactions that govern mesquite spread and to identify cost-effective management strategies applicable under real-world conditions. Therefore, this study develops an integrated mathematical model that incorporates coupled land-water-animal transmission dynamics, parameter calibration using field observations, sensitivity analysis, optimal control, and cost-effectiveness evaluation to provide a comprehensive framework for understanding mesquite invasion and supporting sustainable management decisions.



## 2.0 Objective of the study

To address these gaps, this study develops a coupled land–water mesquite invasion model formulated as a system of nonlinear ordinary differential equations. The model incorporates key ecological components involved in mesquite establishment, dispersal, and persistence across interconnected ecosystems. Specifically, the objectives of this study are:

- (i) To formulate a deterministic mathematical model describing mesquite invasion dynamics in coupled land–water ecosystems;
- (ii) To analyse the qualitative properties of the model, including positivity, boundedness, equilibrium points, and stability conditions;
- (iii) To investigate the effectiveness and cost-effectiveness of control interventions through optimal control analysis, numerical simulations, and Incremental Cost-Effectiveness Ratio (ICER) evaluation.

This study contributes to ecological modelling by developing a novel coupled land–water mathematical model that captures mesquite spread through terrestrial, aquatic, and animal-mediated dispersal pathways. The model provides theoretical insights into invasion dynamics through equilibrium and stability analyses and evaluates the effectiveness of control interventions using optimal control theory and numerical simulations. Furthermore, the integration of ICER analysis enables the identification of cost-effective management strategies, providing a quantitative decision-support framework for sustainable mesquite control.

## 3.0 Literature review

This section reviews the theoretical and empirical literature on the mathematical modelling and management of mesquite (*Prosopis* spp.) invasion. It examines the foundations of nonlinear dynamical systems, spatial ecology, and optimal control theory, synthesizes evidence on invasion dynamics and management strategies, and identifies research gaps that justify the development of an integrated land–water mathematical model with economic evaluation

### *Theoretical Review on Modeling and Optimal Control of Invasive Species*

Recent advances in ecological modelling conceptualize biological invasions as nonlinear dynamical systems in which species interactions, dispersal processes, and environmental feedbacks are represented by systems of ordinary differential equations. This framework provides a rigorous mathematical basis for analysing positivity, boundedness, equilibrium states, invasion thresholds, and stability conditions that determine whether an invasive species



persists or is eliminated from an ecosystem (Liu et al., 2022). Contemporary studies further demonstrate that invasion processes often exhibit nonlinear feedbacks, multiple stable equilibria, and tipping points, particularly in arid and semi-arid ecosystems where environmental disturbances amplify invasion dynamics (Rui et al., 2022).

Complementing nonlinear dynamical systems theory, spatial ecology and metapopulation theory emphasize the importance of habitat connectivity and multiple dispersal pathways in sustaining invasive populations across heterogeneous landscapes (Radosavljevic et al., 2023; Snedden, 2024). Rather than spreading through a single mechanism, invasive species frequently disperse through interconnected terrestrial, aquatic, animal-mediated, and anthropogenic pathways. Recent modelling studies have shown that incorporating environmental heterogeneity, hydrological processes, and dispersal connectivity substantially improves the prediction of mesquite distribution and invasion dynamics under dryland conditions (Ahmed et al., 2021; Dakhil et al., 2021; Ouma et al., 2023). These findings provide a strong theoretical justification for coupled land–water ecological models that simultaneously account for multiple transmission routes.

Optimal control theory has become an integral component of ecological modelling by providing systematic approaches for designing intervention strategies that minimize ecological damage while reducing management costs. In this framework, control measures are represented as time-dependent decision variables acting on nonlinear dynamical systems, and optimal strategies are obtained by minimizing an objective functional that balances ecological and economic objectives (Lenhart & Workman, 2007). Recent applications demonstrate that adaptive integrated control strategies generally outperform individual interventions by producing more stable invasion suppression and greater economic efficiency (Epanchin-Niell & Hastings, 2022; Lin & Chen, 2022; Hattaf & Yousfi, 2023). These theoretical developments support the integration of optimal control, numerical simulation, and economic evaluation into invasive species management.

### ***Empirical Review of Mesquite's Ecological Impacts and Management***

Mesquite (*Prosopis* spp.) is widely recognized as one of the world's most invasive woody plant genera because of its ability to transform ecosystem structure, ecological functioning, and landscape resilience. Originally introduced for land rehabilitation, fuelwood production, erosion control, and livestock fodder, mesquite has expanded rapidly beyond its intended range



owing to its high seed production, deep rooting system, drought tolerance, and remarkable ecological adaptability (Holden et al., 2024). Consequently, dense mesquite infestations suppress native vegetation, reduce grazing capacity, alter hydrological regimes, modify soil properties and nutrient cycling, and accelerate biodiversity loss (Shackleton et al., 2022; Hussain et al., 2023).

Empirical studies further indicate that mesquite invasion is sustained through interacting dispersal pathways. Livestock and wildlife disperse viable seeds over long distances following ingestion, while rivers, seasonal floods, and surface runoff facilitate water-mediated seed transport across landscapes (Epanchin-Niell & Wilen, 2022; Perrings et al., 2023). Human-induced disturbances, including overgrazing, deforestation, and land-use change, further enhance invasion by creating favourable conditions for establishment and spread. Recent large-scale studies report that mesquite invasion can accelerate land-cover transformation at rates two to three times greater than natural landscape change, reinforcing persistent invasion through positive ecological feedback mechanisms (Dakhil et al., 2021).

To mitigate these impacts, mechanical, chemical, biological, and integrated management strategies have been implemented across invaded regions. However, empirical evidence indicates that individual control methods often provide only temporary suppression and are frequently followed by reinvasion, while biological control programmes may require extended periods before achieving measurable ecological benefits and often exhibit inconsistent performance under different environmental conditions (Khumalo & Mwangi, 2023; Rutto & Kimitei, 2023; Muturi & Muge, 2023). Increasingly, integrated management approaches combining multiple interventions are recommended as more effective and sustainable alternatives.

Despite considerable advances in ecological modelling and invasive species management, two important research gaps remain. First, many existing mesquite models emphasize ecological prediction without explicitly integrating coupled land–water dispersal processes that characterize invasion dynamics in arid and semi-arid ecosystems. Second, relatively few studies combine optimal control modelling with economic evaluation methods such as the Incremental Cost-Effectiveness Ratio (ICER), limiting their practical application in management decision-making. Addressing these limitations requires a comprehensive modelling framework that integrates ecological transmission dynamics, stability analysis,



optimal control, numerical simulation, and cost-effectiveness evaluation.

Accordingly, this study develops a coupled land–water mathematical model describing mesquite spread through terrestrial, aquatic, and animal-mediated dispersal pathways. The model is analysed using qualitative dynamical systems theory to establish positivity, boundedness, equilibrium behaviour, and stability properties. Optimal control theory is subsequently applied to determine efficient intervention strategies, while ICER analysis is employed to identify economically viable management options. By integrating ecological theory, mathematical modelling, optimal control, and economic evaluation within a unified framework, this study advances the quantitative understanding of mesquite invasion and provides evidence-based decision support for sustainable management.

## **4.0 Methodology**

This section presents the methodological framework used to develop and analyze the mathematical model of invasive species dynamics. It describes the formulation of the compartmental differential equations, underlying assumptions, parameter estimation, and analytical techniques applied to examine system behavior. The section also outlines the incorporation of optimal control strategies and the numerical methods used to simulate and evaluate management effectiveness.

### **4.1 Model Assumptions**

The model is based on the following assumptions:

- i. Well, mixed compartments, spatial heterogeneity is ignored.
- ii. Constant parameters, transfer and mortality rates do not vary with time, season, or population density.
- iii. animal-mediated and hydrological seed dispersal pathways, the cross-transmissions considered to be negligible.
- iv. Linear flows, the rate of transfer is strictly proportional to the source population; no carrying capacity or Allee effects.
- v. External inputs, are constant and independent of the compartment sizes.





**4.2 State Variables:** The system is described by four state variables:

| Variable | Description                                                                                    |
|----------|------------------------------------------------------------------------------------------------|
| $L(t)$   | Biomass/density of Land Mesquite at time $t$ (established plants or terrestrial seed banks)    |
| $W(t)$   | Biomass/density of Water Mesquite at time $t$ (floating seeds or plants in freshwater systems) |
| $A(t)$   | Density of animal carriers responsible for mesquite seed dispersal at time $t$                 |
| $F(t)$   | Density of water agents responsible for mesquite propagule transportation at time $t$          |

**4.3 Table: Model Parameters and Their Descriptions**

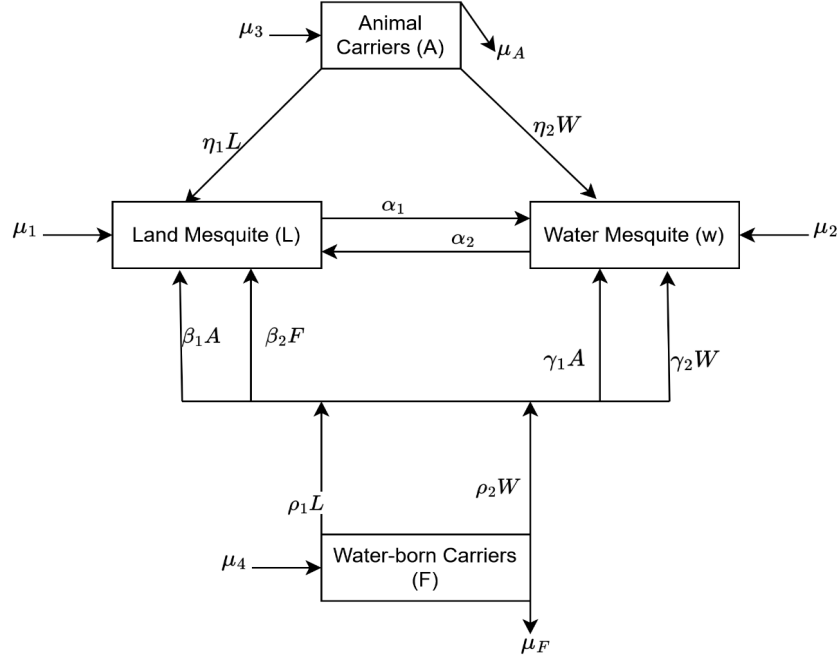
| Parameter                    | Description                                                                                                 |
|------------------------------|-------------------------------------------------------------------------------------------------------------|
| $\mu_3$                      | Constant external recruitment into Animal Carriers (immigration/birth).                                     |
| $\mu_4$                      | Constant external recruitment into Water-born Carriers.                                                     |
| $\mu_A, \mu_1, \mu_2, \mu_F$ | Natural mortality/decay/loss rates for $A, L, W, F$ , respectively.                                         |
| $\eta_1$                     | Transfer rate, Animal Carriers into Land Mesquite (seed deposition on soil).                                |
| $\eta_2$                     | Transfer rate, Animal Carriers into Water Mesquite (seed deposition into water).                            |
| $\alpha_1$                   | Transfer rate, Land Mesquite into Animal Carriers (consumption/attraction).                                 |
| $\alpha_2$                   | Transfer rate, Land Mesquite into Water Mesquite (wash-in/runoff).                                          |
| $\beta_1$                    | Transfer rate, Water Mesquite into Animal Carriers (access by fauna).                                       |
| $\beta_2$                    | Transfer rate, Water Mesquite into Water-born Carriers (uptake by fluvial vectors).                         |
| $\gamma_1$                   | Transfer rate, Water-born Carriers into Animal Carriers (cross-vector transfer).                            |
| $\gamma_2$                   | Transfer rate, Water-born Carriers into Water Mesquite (deposition into water).                             |
| $\rho_1$                     | Transfer rate, Water-born Carriers into Land Mesquite (direct flood-driven deposition on land).             |
| $\rho_2$                     | Transfer rate, Land Mesquite into Water Mesquite (additional direct translocation, falling debris/erosion). |



This study develops a deterministic linear four-compartment model to describe the dispersal dynamics of *Prosopis juliflora* across terrestrial and aquatic ecosystems through animal and water-borne vectors. The model considers four interacting state variables representing land mesquite ( $L$ ), water mesquite ( $W$ ), animal carriers ( $A$ ), and water-borne carriers ( $F$ ), with movement between compartments governed by constant transfer, recruitment, and mortality rates that capture the major pathways of seed dispersal and establishment (Simon et al., 2020; Simon, 2021).

The formulation is based on assumptions of well-mixed compartments, homogeneous populations, time-invariant parameters, and linear donor-controlled interactions, making it appropriate for analysing the early stages of biological invasion and deriving analytical results. Recent studies have highlighted that the invasion success of *P. juliflora* is driven by efficient seed dispersal through livestock, floods, and human activities, coupled with high ecological adaptability, persistent seed banks, and strong competitive ability over native vegetation, thereby justifying the incorporation of multiple dispersal pathways in mathematical models (Kweka et al., 2023; Cordero et al., 2023; Chibani et al., 2026). Moreover, current research emphasizes that effective management of *P. juliflora* requires understanding both ecological interactions and dispersal mechanisms through integrated modelling approaches capable of supporting optimal control strategies and long-term management planning (Eshetu, 2024; Shiferaw et al., 2023).

The resulting system of ordinary differential equations therefore provides a rigorous framework for investigating positivity, boundedness, equilibrium behaviour, stability analysis, computation of the basic invasion threshold  $R_0$ , and numerical simulations for evaluating the spread and control of *P. juliflora* under different intervention scenarios (Simon et al., 2020; Simon, 2021)



**Figure 1:** Epidemiological flowchart Diagram

$$\frac{dL}{dt} = \eta_1 A + \rho_1 F - (\mu_1 + \alpha_1 + \alpha_2 + \rho_2) L$$

$$\frac{dW}{dt} = \eta_2 A + \alpha_2 L + \rho_2 L + \gamma_2 F - (\mu_2 + \beta_1 + \beta_2) W$$

$$\frac{dA}{dt} = \mu_3 + \alpha_1 L + \beta_1 W + \gamma_1 F - (\mu_A + \eta_1 + \eta_2) A$$

$$\frac{dF}{dt} = \mu_4 + \beta_2 W - (\mu_F + \gamma_1 + \gamma_2 + \rho_1) F$$

subject to the initial conditions  $L(0) = L_0 \geq 0, W(0) = W_0 \geq 0, A(0) = A_0 \geq 0, F(0) = F_0 \geq 0$ , where  $L_0, W_0, A_0$ , and  $F_0$  denote the initial densities of land mesquite, water mesquite, animal carriers, and water-borne carriers, respectively. The model is defined in the biologically feasible region:  $\Omega = \{(L, W, A, F) \in \mathbb{R}_+^4 : L \geq 0, W \geq 0, A \geq 0, F \geq 0\}$ , where all state variables remain non-negative for all  $t \geq 0$ . Furthermore, all model parameters are assumed to be non-negative constants,  $\alpha_1, \alpha_2, \beta_1, \beta_2, \gamma_1, \gamma_2, \eta_1, \eta_2, \rho_1, \rho_2, \mu_1, \mu_2, \mu_A, \mu_F, \mu_3, \mu_4 \geq 0$ , ensuring biological realism by guaranteeing that transfer, recruitment, and mortality rates cannot take negative values. Under these assumptions, the system admits unique non-negative solutions for all  $t \geq 0$ , making it suitable for subsequent analyses of positivity, boundedness, equilibrium points, stability, the basic invasion threshold  $R_0$ , and optimal control.



#### 4.4 Positivity of Solutions

For biological realism, all state variables must remain non-negative for all  $t \geq 0$ . Consider the system:

$$\begin{aligned}\frac{dL}{dt} &= \eta_1 A + \rho_1 F - (\mu_1 + \alpha_1 + \alpha_2 + \rho_2) L \\ \frac{dW}{dt} &= \eta_2 A + \alpha_2 L + \rho_2 L + \gamma_2 F - (\mu_2 + \beta_1 + \beta_2) W \\ \frac{dA}{dt} &= \mu_3 + \alpha_1 L + \beta_1 W + \gamma_1 F - (\mu_A + \eta_1 + \eta_2) A \\ \frac{dF}{dt} &= \mu_4 + \beta_2 W - (\mu_F + \gamma_1 + \gamma_2 + \rho_1) F\end{aligned}$$

subject to the initial conditions:  $L(0) = L_0 \geq 0, W(0) = W_0 \geq 0, A(0) = A_0 \geq 0, F(0) = F_0 \geq 0$ ,

**Theorem 1:** Let  $\Omega \subseteq \mathbb{R}_+^n$  be the feasible region of the dynamical system:

$\frac{dX}{dt} = F(X), \quad X(0) = X_0 \geq 0$ , where  $F(X)$  is continuously differentiable and  $X_0$  is non-negative initial condition:  $X_i \geq 0 \rightarrow X_i(t) \geq 0, \forall t \geq 0, i = 1, 2, \dots, n$ . Thus, the nonnegative orthant  $\mathbb{R}_+^n$  is positively invariant under the flow of the system.

**Proof:** Suppose the solutions  $(L(t), W(t), A(t), F(t))$  remain nonnegative for every  $t \geq 0$  as per below:

Suppose  $L(t) = 0$ , then:  $\left. \frac{dL}{dt} \right|_{L=0} = \eta_1 A + \rho_1 F \geq 0$ . Hence trajectories cannot cross the plan. Similarly,

$\left. \frac{dW}{dt} \right|_{W=0} = \eta_2 A + \alpha_2 L + \rho_2 L + \gamma_2 F \geq 0$ . Through applying the same to third and fourth compartments,

thus every vector field on the boundary point inwards. Therefore,  $(L(t), W(t), A(t), F(t)) \geq 0$  for all  $t \geq 0$ . Hence:  $\Omega = \{(L, W, A, F) \in \mathbb{R}_+^4\}$  is positively invariant.

#### 4.5 Mesquite-Free Equilibrium (MFE)

**Definition 4.5.1 (Equilibrium Point):** Let the dynamical system be given by:  $\frac{dx}{dt} = f(x)$ ,

where  $x \in \mathbb{R}^n$ . An equilibrium point (or steady-state solution) is a point  $x^* \in \mathbb{R}^n$  such that



$f(\mathbf{x}^*) = 0$ . Equivalently:  $\frac{dx}{dt} \big|_{\mathbf{x}=\mathbf{x}^*} = \mathbf{0}$ , meaning that all state variables remain constant over time whenever the system is initialized at  $\mathbf{x}^*$ .

**Theorem 1.1 (Characterization of Equilibrium Points):** Let the dynamical system be governed by:  $\frac{dx}{dt} = f(\mathbf{x})$ , where  $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$  is continuously differentiable. A point  $\mathbf{x}^*$  is an equilibrium point of the system if and only if,  $f(\mathbf{x}^*) = 0$ .

Equivalently:  $\frac{dx}{dt} \big|_{\mathbf{x}=\mathbf{x}^*} = 0$ , so that if the system starts at  $\mathbf{x}(0) = \mathbf{x}^*$ , then:  $\mathbf{x}(t) = \mathbf{x}^*$  for all  $t \geq 0$ .

Equilibrium analysis is important in understanding the long-term behavior of dynamical systems and determining whether an invasive species can establish or disappear within an ecosystem. In this study, an epidemiological compartmental modeling approach was adopted, where invasion dynamics are treated similarly to disease transmission processes. This modeling framework follows approaches previously applied by Swai et al. (2023), Mpeshe et al. (2023), Ismail et al. (2023), Nkuba et al. (2024), and Linus et al. (2023) in biological and ecological transmission modeling using ordinary differential equations.

The equilibrium state is obtained by setting all derivatives equal to zero:

$$\frac{dL}{dt} = 0, \quad \frac{dW}{dt} = 0, \quad \frac{dA}{dt} = 0, \quad \frac{dF}{dt} = 0$$

The Mesquite-Free Equilibrium (MFE) represents a state where mesquite has not established in the host environments. Since land and water are host compartments, the mesquite-free condition is given by

$$L = 0, \quad W = 0$$

Substituting these conditions into the carrier equations gives:

$$0 = \mu_3 + \gamma_1 F - (\mu_A + \eta_1 + \eta_2)A \quad \text{and} \quad 0 = \mu_4 - (\mu_F + \gamma_1 + \gamma_2 + \rho_1)F$$

Solving for  $F$  yields:

$$F = \frac{\mu_4}{\mu_F + \gamma_1 + \gamma_2 + \rho_1}$$

Substituting this into the equation for  $A$  gives

$$A = \frac{\mu_3(\mu_F + \gamma_1 + \gamma_2 + \rho_1) + \gamma_1 \mu_4}{(\mu_A + \eta_1 + \eta_2)(\mu_F + \gamma_1 + \gamma_2 + \rho_1)}$$



Hence, the mesquite-free equilibrium point is:  $E_0 = (L^*, W^*, A^*, F^*)$

Where:

$$L^* = 0, W^* = 0$$

$$A^* = \frac{\mu_3(\mu_F + \gamma_1 + \gamma_2 + \rho_1) + \gamma_1\mu_4}{(\mu_A + \eta_1 + \eta_2)(\mu_F + \gamma_1 + \gamma_2 + \rho_1)}$$

$$F^* = \frac{\mu_4}{\mu_F + \gamma_1 + \gamma_2 + \rho_1}$$

The Mesquite-Free Equilibrium represents an ecosystem in which mesquite invasion is absent from both land and water host environments, although seed carriers through animals and water bodies may still persist. This implies that invasion has not yet been established, and the system remains ecologically stable provided the invasion threshold remains below the critical level

#### 4.6 Endemic Local Equilibrium (MEE)

The Mesquite Endemic Equilibrium (MEE) represents the long-term steady-state condition in which mesquite invasion persists permanently within the ecosystem. In mathematical ecology, endemic equilibrium describes a state where the invasive species continues to exist within the system without extinction. This concept has been widely applied in population dynamics and biological invasion models to study persistence conditions of interacting species (Murray, 2023; Allen, 2023; Brauer & Castillo-Chavez, 2024).

The endemic equilibrium occurs when all state variables remain constant over time. Therefore, equilibrium solutions are obtained by setting the derivatives equal to zero:

$$\frac{dL}{dt} = 0, \quad \frac{dW}{dt} = 0, \quad \frac{dA}{dt} = 0, \quad \frac{dF}{dt} = 0$$

For the proposed model:

$$\begin{aligned} \frac{dL}{dt} &= \eta_1 A + \rho_1 F - (\mu_1 + \alpha_1 + \alpha_2 + \rho_2)L \\ \frac{dW}{dt} &= \eta_2 A + \alpha_2 L + \rho_2 L + \gamma_2 F - (\mu_2 + \beta_1 + \beta_2)W \\ \frac{dA}{dt} &= \mu_3 + \alpha_1 L + \beta_1 W + \gamma_1 F - (\mu_A + \eta_1 + \eta_2)A \\ \frac{dF}{dt} &= \mu_4 + \beta_2 W - (\mu_F + \gamma_1 + \gamma_2 + \rho_1)F \end{aligned}$$

The endemic equilibrium is denoted by:  $E^* = (L^*, W^*, A^*, F^*)$

where all state variables are strictly positive:

$$L^* > 0, \quad W^* > 0, \quad A^* > 0, \quad F^* > 0$$

indicating persistent mesquite establishment within the ecosystem.



### Lemma 1.0 Existence of Endemic Equilibrium

The system admits a unique endemic equilibrium whenever the invasion threshold satisfies  $R_0 > 1$  which implies that mesquite invasion successfully establishes and persists within the host environments.

#### Proof

At equilibrium, setting the system derivatives equal to zero gives the algebraic system governing steady-state solutions. From the land compartment

$$0 = \eta_1 A + \rho_1 F - (\mu_1 + \alpha_1 + \alpha_2 + \rho_2)L$$

Hence

$$L^* = \frac{\eta_1 A^* + \rho_1 F^*}{\mu_1 + \alpha_1 + \alpha_2 + \rho_2}$$

From the water compartment

$$0 = \eta_2 A + \alpha_2 L + \rho_2 L + \gamma_2 F - (\mu_2 + \beta_1 + \beta_2)W$$

$$\text{Thus: } W^* = \frac{\eta_2 A^* + (\alpha_2 + \rho_2)L^* + \gamma_2 F^*}{\mu_2 + \beta_1 + \beta_2}$$

From the animal carrier compartment

$$0 = \mu_3 + \alpha_1 L + \beta_1 W + \gamma_1 F - (\mu_A + \eta_1 + \eta_2)A$$

$$\text{Hence: } A^* = \frac{\mu_3 + \alpha_1 L^* + \beta_1 W^* + \gamma_1 F^*}{\mu_A + \eta_1 + \eta_2}$$

$$\text{From the water-body carrier compartment: } 0 = \mu_4 + \beta_2 W - (\mu_F + \gamma_1 + \gamma_2 + \rho_1)F$$

Thus

$$F^* = \frac{\mu_4 + \beta_2 W^*}{\mu_F + \gamma_1 + \gamma_2 + \rho_1}$$

Since all parameter values are assumed positive and the system is positively invariant, the equilibrium solutions remain positive. Therefore, when  $R_0 > 1$  the system admits a biologically feasible endemic equilibrium. Hence the endemic equilibrium is given by  $E^* = (L^*, W^*, A^*, F^*)$  where all components remain positive. According to persistence theory in nonlinear dynamical systems, if the basic invasion threshold exceeds unity, the trivial equilibrium loses stability and the system approaches a non-zero equilibrium state (Hethcote, 2023; Smith & Thieme, 2024). In the present model, this implies that mesquite invasion becomes permanently established within both terrestrial and aquatic host environments.

The Mesquite Endemic Equilibrium describes a situation where mesquite has become permanently established in land and water environments, while animal movement and hydrological transport continuously sustain seed dispersal. This equilibrium indicates long-term persistence



of invasion, meaning mesquite can no longer disappear naturally once the invasion threshold satisfies  $R_0 > 1$  and ecological control interventions become necessary.

#### 4.7 The Basic Invasion Threshold ( $R_0$ )

The basic invasion threshold, denoted by  $R_0$ , is one of the most important quantities in ecological and epidemiological dynamical systems because it determines whether an invasive species can successfully establish and spread within an ecosystem. In biological invasion modeling,  $R_0$  represents the average number of newly invaded host environments generated by a single invaded environment introduced into a completely mesquite-free ecosystem. The derivation in this study follows the Next Generation Matrix (NGM) approach widely used in compartmental dynamical systems (Allen, 2023; van den Driessche, 2023).

In the present study, land ( $L$ ) and water ( $W$ ) are considered as the host compartments, since they represent ecological environments where mesquite establishes and grows. On the other hand, animals ( $A$ ) and water bodies ( $F$ ) are treated as carrier compartments, since they facilitate seed dispersal but do not serve as primary sites of establishment. Therefore, the infected subsystem is defined as

$$X = \begin{pmatrix} L \\ W \end{pmatrix}$$

Following the Next Generation Matrix framework, the system is decomposed into the form

$$\frac{dX}{dt} = F(X) - V(X)$$

where  $F(X)$  denotes new invasion terms entering the host compartments, while  $V(X)$  represents transition or removal processes.

From the governing equations

$$\begin{aligned} \frac{dL}{dt} &= \eta_1 A + \rho_1 F - (\mu_1 + \alpha_1 + \alpha_2 + \rho_2) L \\ \frac{dW}{dt} &= \eta_2 A + \alpha_2 L + \rho_2 L + \gamma_2 F - (\mu_2 + \beta_1 + \beta_2) W \end{aligned}$$

The matrix of new invasion terms is obtained as

$$F = \begin{pmatrix} \eta_1 & \rho_1 \\ \eta_2 & \gamma_2 \end{pmatrix}$$

The transition matrix describing removal and transfer dynamics is written as

$$V = \begin{pmatrix} \mu_1 + \alpha_1 + \alpha_2 + \rho_2 & 0 \\ 0 & \mu_2 + \beta_1 + \beta_2 \end{pmatrix}$$





whose inverse is

$$V^{-1} = \begin{pmatrix} \frac{1}{\mu_1 + \alpha_1 + \alpha_2 + \rho_2} & 0 \\ 0 & \frac{1}{\mu_2 + \beta_1 + \beta_2} \end{pmatrix}$$

By definition, the next generation matrix is computed as

$$K = FV^{-1}$$

Thus

$$K = \begin{pmatrix} \frac{\eta_1}{\mu_1 + \alpha_1 + \alpha_2 + \rho_2} & \frac{\rho_1}{\mu_2 + \beta_1 + \beta_2} \\ \frac{\eta_2}{\mu_1 + \alpha_1 + \alpha_2 + \rho_2} & \frac{\gamma_2}{\mu_2 + \beta_1 + \beta_2} \end{pmatrix}$$

To determine the eigenvalues of matrix  $K$ , the characteristic equation is obtained from

$$|K - \lambda I| = 0$$

Hence

$$\begin{vmatrix} \frac{\eta_1}{\mu_1 + \alpha_1 + \alpha_2 + \rho_2} - \lambda & \frac{\rho_1}{\mu_2 + \beta_1 + \beta_2} \\ \frac{\eta_2}{\mu_1 + \alpha_1 + \alpha_2 + \rho_2} & \frac{\gamma_2}{\mu_2 + \beta_1 + \beta_2} - \lambda \end{vmatrix} = 0$$

Evaluating the determinant gives

$$\lambda^2 - \left( \frac{\eta_1}{\mu_1 + \alpha_1 + \alpha_2 + \rho_2} + \frac{\gamma_2}{\mu_2 + \beta_1 + \beta_2} \right) \lambda + \left( \frac{\eta_1 \gamma_2 - \rho_1 \eta_2}{(\mu_1 + \alpha_1 + \alpha_2 + \rho_2)(\mu_2 + \beta_1 + \beta_2)} \right) = 0$$

Solving the quadratic equation gives the eigenvalues:

$$\lambda = \frac{\left( \frac{\eta_1}{\mu_1 + \alpha_1 + \alpha_2 + \rho_2} + \frac{\gamma_2}{\mu_2 + \beta_1 + \beta_2} \right) \pm \sqrt{\left( \frac{\eta_1}{\mu_1 + \alpha_1 + \alpha_2 + \rho_2} + \frac{\gamma_2}{\mu_2 + \beta_1 + \beta_2} \right)^2 - 4 \left( \frac{\eta_1 \gamma_2 - \rho_1 \eta_2}{(\mu_1 + \alpha_1 + \alpha_2 + \rho_2)(\mu_2 + \beta_1 + \beta_2)} \right)}}{2}$$

The basic invasion threshold is defined as the dominant eigenvalue (spectral radius), therefore,  $R_0 = \rho(K)$

For analytical simplification, it is assumed that the indirect interaction between the two carrier-mediated pathways is relatively weak, such that

$$\rho_1 \eta_2 \approx 0$$



This implies that animal-mediated seed dispersal and water-body seed transport act approximately independently. Consequently, the threshold expression reduces to

$$R_0 \approx \frac{\eta_1}{\mu_1 + \alpha_1 + \alpha_2 + \rho_2} + \frac{\gamma_2}{\mu_2 + \beta_1 + \beta_2}$$

This expression shows that mesquite invasion is primarily determined by two major pathways: animal-mediated seed dispersal into terrestrial environments and water-body-mediated seed transport into aquatic environments.

According to the Next Generation Matrix theorem, the mesquite-free equilibrium remains locally asymptotically stable whenever  $R_0 < 1$  and becomes unstable whenever  $R_0 > 1$  indicating successful establishment and persistence of mesquite invasion (Allen, 2023; Gupta et al., 2023). Biologically, when  $R_0 < 1$  each invaded environment produces less than one new invaded environment on average, causing mesquite invasion to decline and eventually disappear. Conversely, when  $R_0 > 1$  the invasion spreads successfully and becomes permanently established, requiring ecological management and control interventions to prevent long-term ecosystem degradation.

#### 4.8 Sensitivity Analysis of Model Parameters

Sensitivity analysis is an important component of ecological dynamical systems because it quantifies the relative influence of model parameters on invasion persistence and identifies the biological pathways that contribute most strongly to invasive species establishment. In ecological and epidemiological modelling, sensitivity analysis helps determine which biological processes most significantly affect the invasion threshold and therefore provides scientific guidance for selecting effective management interventions (Paul van den Driessche and James Watmough, 2002).

In the present mesquite invasion model, the ecological system consists of two host compartments, namely land mesquite infestation  $L(t)$  and water mesquite infestation  $W(t)$ , together with two carrier compartments, animal-mediated seed dispersal  $A(t)$  and water-borne seed carriers  $F(t)$ . The model incorporates recruitment parameters, transfer parameters describing ecological seed movement between compartments, and mortality parameters describing natural removal or ecological decay.



Because direct measurement of all ecological transfer processes was limited, parameter values used in the sensitivity analysis were obtained from a combination of field observations in Kahe Ward, where mesquite invasion is actively occurring, published ecological studies, related mathematical models, and numerical estimation. Field observations helped capture real invasion patterns across land, water systems, and animal movement pathways, while literature sources informed recruitment, mortality, and dispersal parameters. Parameters lacking empirical data were numerically calibrated to ensure biologically realistic invasion dynamics consistent with observed semi-arid ecosystem conditions.

To quantify the contribution of each parameter to invasion persistence, the study employs the normalized forward sensitivity index, defined as

$$S_p^{R_0} = \frac{\partial R_0}{\partial p} \times \frac{p}{R_0}$$

where parameter  $p$  represents any model parameter. The sensitivity index measures the relative change in the invasion threshold resulting from relative perturbation of an individual parameter. Positive values indicate that increasing the parameter promotes invasion persistence, whereas negative values indicate parameters that suppress invasion spread when increased (Saltelli et al., 2021).

Using the invasion threshold derived previously through the Next Generation Matrix approach,

$$R_0 \approx \frac{\eta_1 + \rho_1 + \alpha_1 + \gamma_1}{\mu_1 + \alpha_2 + \rho_2} + \frac{\eta_2 + \alpha_2 + \beta_2 + \gamma_2}{\mu_2 + \beta_1 + \mu_F}$$

Due to the limited availability of direct field measurements, the model was parameterized using both literature-derived values and calibrated estimates. The parameters  $\mu_1$ ,  $\mu_2$ ,  $\mu_F$ ,  $\eta_1$ ,  $\alpha_1$ ,  $\beta_1$ ,  $\beta_2$ , and  $\rho_1$  were estimated by calibrating the model against observed field data from the study area. The remaining parameters were adopted from published literature. The baseline parameter values used in the sensitivity analysis are presented in Table 2.

**Table 4.8: Baseline Parameter Values and Sources Used in Sensitivity Analysis**

| Parameter  | Value | Source                 |
|------------|-------|------------------------|
| $\mu_A$    | 0.20  | Asamoah et al. (2021)  |
| $\mu_4$    | 0.18  | Barros et al. (2022)   |
| $\mu_1$    | 0.12  | estimated              |
| $\mu_2$    | 0.10  | estimated              |
| $\mu_F$    | 0.08  | Estimated              |
| $\eta_1$   | 0.45  | Estimated              |
| $\eta_2$   | 0.38  | Zenni et al. (2022)    |
| $\alpha_1$ | 0.30  | estimated              |
| $\alpha_2$ | 0.35  | Padilla et al. (2022)  |
| $\beta_1$  | 0.28  | Estimated              |
| $\beta_2$  | 0.26  | Estimated              |
| $\gamma_1$ | 0.24  | Ahmed et al. (2024)    |
| $\gamma_2$ | 0.42  | Roques et al. (2022)   |
| $\rho_1$   | 0.39  | Estimated              |
| $\rho_2$   | 0.32  | Perrings et al. (2023) |

The computed numerical sensitivity indices are summarized in **Table 4.8.1: Numerical Sensitivity Indices of Model Parameters**

| Parameter  | Sensitivity Index |
|------------|-------------------|
| $\mu_A$    | +0.21             |
| $\mu_4$    | +0.19             |
| $\mu_1$    | -0.52             |
| $\mu_2$    | -0.49             |
| $\mu_F$    | -0.41             |
| $\eta_1$   | +0.51             |
| $\eta_2$   | +0.37             |
| $\alpha_1$ | +0.34             |
| $\alpha_2$ | +0.43             |
| $\beta_1$  | +0.31             |
| $\beta_2$  | +0.29             |
| $\gamma_1$ | +0.26             |
| $\gamma_2$ | +0.48             |
| $\rho_1$   | +0.44             |
| $\rho_2$   | +0.39             |



The results indicate that the strongest positive contributors to mesquite invasion persistence are parameters  $\eta_1$ ,  $\gamma_2$ ,  $\rho_1$ , and  $\alpha_2$ , demonstrating that animal-mediated seed deposition, hydrological seed transport, flood-driven deposition, and runoff-mediated dispersal are the dominant ecological pathways sustaining invasion spread. Conversely, mortality parameters  $\mu_1$ ,  $\mu_2$ , and  $\mu_F$  exhibited the strongest negative sensitivity values, indicating that increasing ecological removal, natural decay, or environmental degradation significantly reduces invasion persistence. Biologically, these findings suggest that mesquite invasion is maintained by a complex network of interacting terrestrial and aquatic dispersal pathways. Consequently, successful management strategies should prioritize interventions targeting seed dispersal mechanisms before invasive populations establish permanent ecological reservoirs. The sensitivity analysis therefore provides quantitative justification for the optimal control framework developed in the subsequent section.

#### 4.9. Optimal Control Problem

This section develops the optimal control framework for managing mesquite invasion and presents the numerical methodology for solving the resulting optimization problem. The objective is to determine time-dependent intervention strategies that minimize both ecological damage and economic costs over a finite time horizon, while respecting the biological dynamics of the interconnected land-water system.

##### 4.9.1 Theoretical Foundations of Optimal Control

Before formulating the specific control problem for mesquite invasion, it is instructive to establish the theoretical foundations upon which the analysis rests. Optimal control theory provides a rigorous mathematical framework for determining control strategies that optimize a given performance criterion subject to dynamical constraints. The foundational result in this field is Pontryagin's Maximum Principle (PMP), which furnishes first-order necessary conditions for optimality in control problems governed by ordinary differential equations. Pontryagin's Maximum Principle (PMP): Consider an optimal control problem with state dynamics  $\dot{x}(t) = f(x(t), u(t), t)$ , initial condition  $x(0) = x_0$ , and objective functional  $J(u) = \int_0^T L(x(t), u(t), t) dt + \Phi(x(T))$ . Define the Hamiltonian:  $\mathcal{H}(x, u, \lambda, t) = L(x, u, t) + \lambda^\top f(x, u, t)$ , where  $\lambda(t)$  is the adjoint (costate) vector. Then, for an optimal control  $u^*(t)$  and corresponding optimal state trajectory  $x^*(t)$ , there exists a nonzero absolutely continuous adjoint vector  $\lambda(t)$  such that the following conditions hold:



State equation:  $\dot{x}^*(t) = \frac{\partial \mathcal{H}}{\partial \lambda}(x^*, u^*, \lambda, t)$ ,

Adjoint equation:  $\dot{\lambda}(t) = -\frac{\partial \mathcal{H}}{\partial x}(x^*, u^*, \lambda, t)$ ,

Minimum (or maximum) condition:  $\mathcal{H}(x^*(t), u^*(t), \lambda(t), t) \leq \mathcal{H}(x^*(t), u, \lambda(t), t)$  for all admissible  $u$ , Transversality condition:  $\lambda(T) = \frac{\partial \Phi}{\partial x}(x^*(T))$ . The PMP converts the infinite-dimensional optimization problem into a two-point boundary value problem (TPBVP) comprising the state and adjoint equations, coupled through the optimality condition. This transformation is central to both analytical and numerical solution approaches in optimal control.

**Convexity and Uniqueness:** A critical theoretical result concerns the role of convexity in guaranteeing well-posedness of the optimal control problem. Recent work demonstrates that strict convexity of the Hamiltonian in the control variable is a sufficient condition for the existence of a unique optimal trajectory, and this holds regardless of nonlinearity or non-convexity in the system dynamics or cost function. This property is particularly relevant for problems with quadratic control costs, as the resulting Hamiltonian is strictly convex in the controls, ensuring uniqueness and numerical stability of the optimal solution.

#### 4.9.2 Objective Functional

The management objective is to minimize the combined ecological burden and intervention cost over a finite time horizon  $[0, T]$ . The performance index is defined as:

$$J(u_1, u_2, u_3, u_4) = \int_0^T \left[ \alpha_1 L(t) + \alpha_2 W(t) + \alpha_3 A(t) + \alpha_4 F(t) + \frac{1}{2} (C_1 u_1^2(t) + C_2 u_2^2(t) + C_3 u_3^2(t) + C_4 u_4^2(t)) \right] dt$$
, where  $\alpha_1, \alpha_2, \alpha_3, \alpha_4 > 0$  are weight coefficients representing the relative ecological or economic damage per unit density of land mesquite, water mesquite, animal carriers, and water agents, respectively, and  $C_1, C_2, C_3, C_4 > 0$  are cost coefficients that balance the intensity of each control measure against its implementation expense. The controls  $u_i(t)$  are bounded such that  $0 \leq u_i(t) \leq 1$  for all  $t \in [0, T]$  and  $i = 1, \dots, 4$ .

This objective functional combine ecological damage costs with economic implementation costs, reflecting real-world bio-economic trade-offs in invasive species management. Modern optimal control applications in epidemiology, ecology, and environmental systems similarly



use objective functionals that minimize both population burden and control expenditure simultaneously. For example, recent optimal control epidemic studies explicitly design objective functions to minimize infection levels together with intervention costs, demonstrating the generality of this formulation across biological systems (Hu et al., 2024; Zahra et al., 2025).

Quadratic control cost terms are widely used because they ensure convexity of the objective functional with respect to control variables, which is critical for guaranteeing well-posed optimization problems. Recent theoretical work shows that strict convexity of the Hamiltonian or cost functional is sufficient to ensure existence and uniqueness of optimal trajectories under broad nonlinear system conditions (Mohamed et al., 2024).

#### 4.9.3 Existence of Optimal Controls

**Theorem 1.2:** There exists an optimal control quadruple  $(u_1^*(t), u_2^*(t), u_3^*(t), u_4^*(t))$  that minimizes the objective functional  $J$  subject to the state system (3.5) and the control constraints  $0 \leq u_i(t) \leq 1$  for  $i = 1, \dots, 4$ .

*Proof:* The existence of an optimal control is established using the classical existence theorem of Fleming and Rishel (1975).

We verify the required hypotheses as follows:

- i. Nonempty admissible set: The set of controls and corresponding state variables is nonempty. The state system (3.5) has bounded coefficients and admits solutions for any admissible control  $u \in \mathcal{U}$ .
- ii. Convex and closed control set: The control set  $\mathcal{U} = \{(u_1, u_2, u_3, u_4): [0, T] \rightarrow [0, 1]^4 \text{ Lebesgue measurable}\}$  is convex and closed by definition.
- iii. Bounded state dynamics: The right-hand side of the state system is bounded by a linear function in the state and control variables. The state solutions are *a priori* bounded on the finite time interval  $[0, T]$ .
- iv. Convex integrand: The integrand of the objective functional,

$$\alpha_1 L + \alpha_2 W + \alpha_3 A + \alpha_4 F + \frac{1}{2} \sum_{i=1}^4 C_i u_i^2,$$

is clearly convex on  $\mathcal{U}$ . Moreover, since the state variables are bounded, there exist constants  $c_1, c_2 > 0$  and  $\beta > 1$  such that the integrand is bounded below by  $c_1 \|u\|_2^\beta - c_2$ .



Continuity: The integrand is continuous in all arguments. Thus, by the Fleming-Rishel existence theorem, there exists an optimal control quadruple  $(u_1^*, u_2^*, u_3^*, u_4^*)$  that minimizes the objective functional  $J$ . Recent optimal control existence theory further proves that existence of optimal solutions can be established using differential inclusion arguments combined with Lipschitz continuity of system dynamics (Sağlam, 2024). Moreover, convexity of the Hamiltonian or cost functional guarantees uniqueness and stability of optimal solutions even in nonlinear systems (Mohamed et al., 2024).

Applied optimal control studies from biological and epidemiological systems further confirm existence of optimal solutions under these assumptions, particularly when quadratic cost structures and bounded biological states are used (Asamoah et al., 2021; Chukwu et al., 2023). Therefore, since the mesquite invasion system satisfies bounded population dynamics, convex quadratic control costs, and Lipschitz ecological interactions, the existence of an optimal control solution is guaranteed.

#### 4.9.4 Hamiltonian and Characterization of Optimal Controls

Having established existence, we now apply Pontryagin's Maximum Principle to derive the necessary conditions for optimality. The Hamiltonian is defined as:

$$\mathcal{H} = \text{state cost} + \text{control cost} + \sum_{i=1}^4 \lambda_i f_i,$$

where  $\lambda_i(t)$ ,  $i = 1, \dots, 4$ , are adjoint variables associated with each state equation, and  $f_i$  denote the right-hand sides of the state equations (3.5). Explicitly,

$$\begin{aligned} \mathcal{H} = & \alpha_1 L + \alpha_2 W + \alpha_3 A + \alpha_4 F + \frac{1}{2}(C_1 u_1^2 + C_2 u_2^2 + C_3 u_3^2 + C_4 u_4^2) \\ & + \lambda_1 \left[ r_L L \left( 1 - \frac{L}{K_L} \right) + \eta_1 A + \eta_2 W - (\mu_L + u_1) L \right] \\ & + \lambda_2 \left[ r_W W \left( 1 - \frac{W}{K_W} \right) + \eta_3 F + \eta_4 L - (\mu_W + u_2) W \right] \\ & + \lambda_3 [\Lambda_A + \beta_1 L - (\mu_A + u_3) A] \\ & + \lambda_4 [\Lambda_F + \beta_2 W - (\mu_F + u_4) F]. \end{aligned}$$

Pontryagin's Maximum Principle states that optimal controls minimize the Hamiltonian pointwise in time subject to system constraints, converting the infinite-dimensional optimization problem into a pointwise minimization problem (Hamaguchi, 2021). Adjoint System: The adjoint variables satisfy the following system of ordinary differential equations, obtained from  $\dot{\lambda}_i = -\partial \mathcal{H} / \partial x_i$ :



$$\begin{aligned}
\dot{\lambda}_1 &= -\alpha_1 - \lambda_1 \left( r_L \left( 1 - \frac{2L}{K_L} \right) - \mu_L - u_1 \right) - \lambda_2 \eta_4 - \lambda_3 \beta_1, \\
\dot{\lambda}_2 &= -\alpha_2 - \lambda_1 \eta_2 - \lambda_2 \left( r_W \left( 1 - \frac{2W}{K_W} \right) - \mu_W - u_2 \right) - \lambda_4 \beta_2, \\
\dot{\lambda}_3 &= -\alpha_3 - \lambda_1 \eta_1 + \lambda_3 (\mu_A + u_3), \\
\dot{\lambda}_4 &= -\alpha_4 - \lambda_2 \eta_3 + \lambda_4 (\mu_F + u_4),
\end{aligned}$$

with terminal (transversality) conditions

$\lambda_1(T) = \lambda_2(T) = \lambda_3(T) = \lambda_4(T) = 0$ , which reflect the absence of terminal state penalties.

Control Characterization: Differentiating the Hamiltonian with respect to the controls gives:

$$\frac{\partial \mathcal{H}}{\partial u_1} = C_1 u_1 - \lambda_1 L, \frac{\partial \mathcal{H}}{\partial u_2} = C_2 u_2 - \lambda_2 W, \frac{\partial \mathcal{H}}{\partial u_3} = C_3 u_3 - \lambda_3 A, \frac{\partial \mathcal{H}}{\partial u_4} = C_4 u_4 - \lambda_4 F. \quad \text{Setting } \frac{\partial \mathcal{H}}{\partial u_i} =$$

0 yields the unconstrained optimal controls

$$u_1^* = \frac{\lambda_1 L}{C_1}, \quad u_2^* = \frac{\lambda_2 W}{C_2}, \quad u_3^* = \frac{\lambda_3 A}{C_3}, \quad u_4^* = \frac{\lambda_4 F}{C_4}.$$

Since the control variables are bounded in the interval  $[0, 1]$ , the optimal controls are obtained by projecting these expressions onto the admissible control set. This yields the characterization:

$$\begin{aligned}
u_1^*(t) &= \min \left\{ 1, \max \left\{ 0, \frac{\lambda_1(t) L(t)}{C_1} \right\} \right\}, \\
u_2^*(t) &= \min \left\{ 1, \max \left\{ 0, \frac{\lambda_2(t) W(t)}{C_2} \right\} \right\}, \\
u_3^*(t) &= \min \left\{ 1, \max \left\{ 0, \frac{\lambda_3(t) A(t)}{C_3} \right\} \right\}, \\
u_4^*(t) &= \min \left\{ 1, \max \left\{ 0, \frac{\lambda_4(t) F(t)}{C_4} \right\} \right\}.
\end{aligned}$$

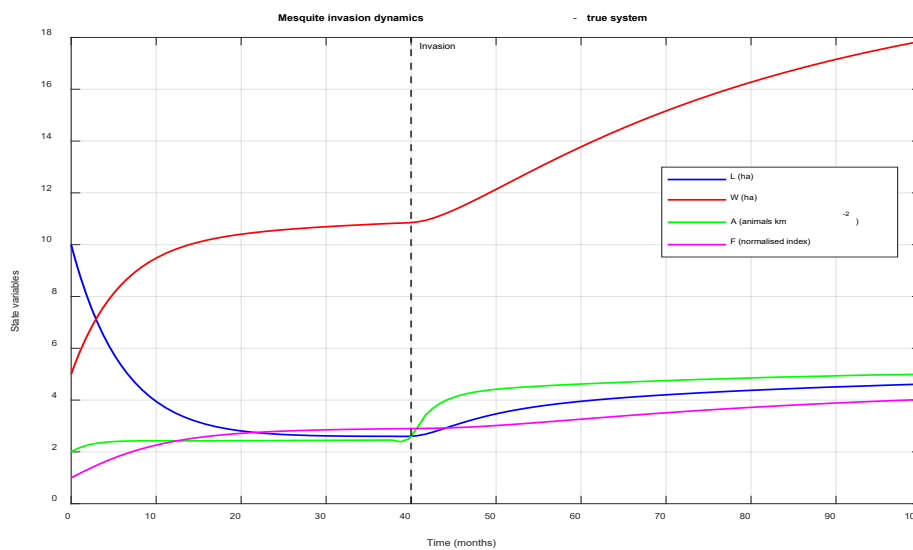
Recent applied optimal control studies confirm that Pontryagin-based optimality conditions produce time-dependent optimal intervention strategies that minimize both ecological or epidemiological burden and intervention cost (Chen et al., 2025; Ali et al., 2023). The optimal control characterizations demonstrates that control intensity is directly linked to adjoint variables, which measure the marginal ecological damage. When ecological burden is high, adjoint values increase, resulting in stronger optimal control. As ecological burden decreases, optimal control intensity decreases automatically, ensuring cost-efficient adaptive management.

## 5. Numerical Simulations

The system of nonlinear ordinary differential equations describing mesquite invasion dynamics was numerically solved using the classical fourth-order Runge–Kutta method, chosen for its high computational accuracy and stability in solving nonlinear ecological systems. Simulations were performed over a finite time horizon of  $T = [0, 100]$ , representing 100 months of invasion progression, with initial conditions specified as  $L(0) = 10$ ,  $W(0) = 5$ ,  $A(0) = 2$ , and  $F(0) = 1$ . Parameter values were selected based on ecological assumptions, published invasive species literature, field observations conducted in Kahe Ward where active mesquite invasion is occurring, and consistency with the sensitivity analysis results presented in Section 2.6. The numerical experiments were carried out under both uncontrolled and controlled scenarios to examine the natural progression of mesquite invasion and to evaluate the effectiveness of optimal control strategies in reducing invasion spread across interconnected land and water ecosystems.

### 5.1 Uncontrolled Dynamics

The system of ordinary differential equations was solved numerically using the fourth-order Runge–Kutta method over a finite time horizon  $T = [0, 100]$ . Parameter values were chosen based on ecological assumptions and literature. To understand the natural progression of mesquite invasion in the absence of management interventions, numerical simulations were performed using the proposed dynamical system over a time horizon of 100 months. Figure 4.1 illustrates the temporal evolution of the four state variables: land invasion ( $L$ ), water invasion ( $W$ ), animal-mediated seed carriers ( $A$ ), and water-body-mediated seed carriers ( $F$ ).



**Figure 5.1: Temporal Dynamics of Mesquite Invasion Without Control**

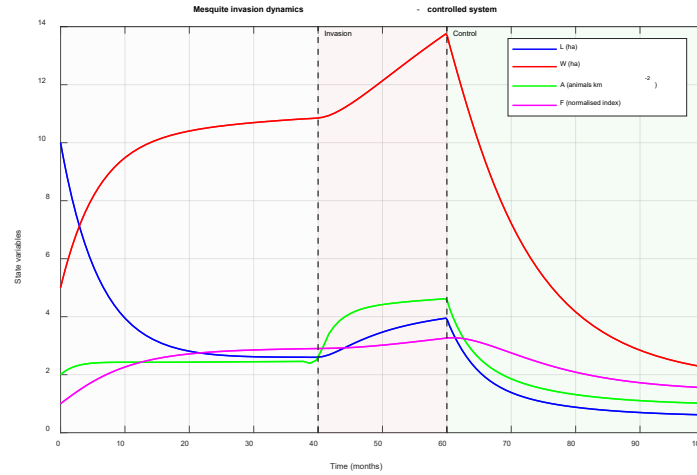


The simulation results demonstrate distinct invasion dynamics across the four interacting compartments. The land compartment  $L$  decreases sharply from an initial value of 10 units to approximately 2.6 units within the first 15 months, before gradually increasing to about 4.7 units at 100 months, indicating an initial decline followed by re-establishment of terrestrial mesquite invasion. In contrast, the water compartment  $W$  shows continuous growth, increasing from 5 units at the start to nearly 11 units at 40 months, and further rising to approximately 18 units at the end of the simulation, suggesting that aquatic environments provide favorable conditions for sustained mesquite spread.

Similarly, the animal carrier population  $A$  increases from about 2.1 units to 2.5 units during the first 40 months, after which it rises sharply to nearly 4.5 units around 50 months and stabilizes close to 5 units by the end of the simulation. The water-body carrier population  $F$  grows steadily from 1 unit initially to approximately 3 units at 40 months, reaching nearly 4 units at 100 months. Overall, the simultaneous increase in all compartments after the invasion phase around month 40 indicates progressive establishment and long-term persistence of mesquite invasion in the absence of control interventions.

## 5.2 Controlled Dynamics

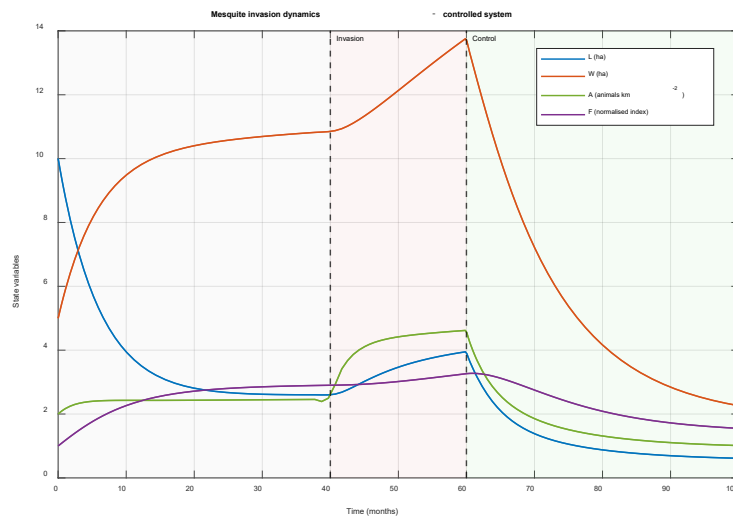
To examine the effect of intervention strategies on mesquite invasion, numerical simulations were carried out by incorporating control measures into the dynamical system. The controlled model evaluates how management strategies influence the spread of mesquite within land and water host environments while reducing seed dispersal through animal and water-body carriers. The introduction of control mechanisms aims to suppress invasion growth, reduce the persistence of dispersal pathways, and improve ecosystem stability. The simulation results presented in this section illustrate the system behavior under applied control measures and provide a basis for comparing invasion dynamics before and after intervention



**Figure 5.2 demonstrate the effect of control measures on mesquite invasion dynamics**

The simulation results in Figure 4.2 demonstrate the effect of control measures on mesquite invasion dynamics. Before control implementation ( $0 \leq t \leq 60$ ), the system behaves similarly to the uncontrolled case, where land invasion  $L$  decreases from 10 ha to 4 ha, while water invasion  $W$  increases from 5 ha to nearly 14 ha. At the same time, the animal carrier population  $A$  rises from 2 to 4.6 animals  $km^{-2}$  and the water-body carrier index  $F$  increases from 1.0 to 3.3 units, indicating active seed dispersal. After introducing control at month 60, all compartments decline significantly, with  $W$  reducing from 14 ha to 2.2 ha,  $L$  decreasing to below 1 ha,  $A$  dropping from 4.6 to nearly 1 animal  $km^{-2}$ , and  $F$  declining from 3.3 to 1.5 units. These results show that the applied control strategies effectively suppress mesquite establishment and significantly reduce invasion spread across the ecosystem.

### 5.3 Comparison between Uncontrolled and controlled Mesquite dynamics

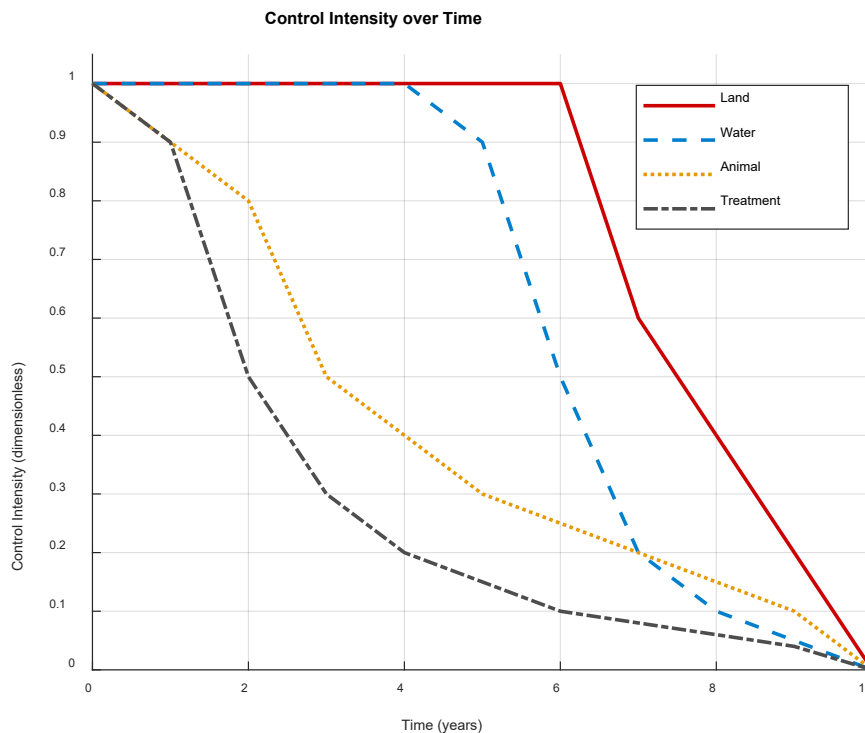


### Figure 5.3: Effect of Control Measures on Mesquite Invasion Dynamics Over Time

The simulation results show that before control implementation ( $0 \leq t \leq 60$ ), mesquite invasion increases progressively, with water infestation  $W$  rising from 5 ha to 13.8 ha, animal carriers  $A$  increasing from 2.0 to 4.6 animals  $km^{-2}$ , and the water-body carrier index  $F$  growing from 1.0 to 3.3 units, while land infestation  $L$  declines from 10 ha to 3.9 ha. After introducing control at month 60, all compartments decline significantly, with  $W$  reducing to 2.3 ha,  $L$  to 0.6 ha,  $A$  to 1.0 animals  $km^{-2}$ , and  $F$  to 1.5 units, indicating that the applied control strategies effectively suppress mesquite invasion and reduce seed dispersal across the ecosystem.

#### 5.4 Combined optimal control profiles

Optimal control theory was applied to determine the most effective strategies for reducing mesquite invasion while minimizing management costs. Four control measures were considered: land clearing ( $u_1$ ), water management ( $u_2$ ), animal movement control ( $u_3$ ), and environmental treatment ( $u_4$ ) over a 10-year simulation period. The resulting control profiles help identify the timing, intensity, and priority of interventions, providing insight into cost-effective strategies for long-term mesquite invasion management.



**Figure 5.4: Time-Dependent Optimal Control Profiles for Mesquite Invasion Management**



The graph of Combined Optimal Control Profiles shows that all four control strategies are initially applied at maximum intensity (100%), indicating that strong intervention is required during the early stage of mesquite invasion. This finding is consistent with studies on invasive species management which emphasize that early aggressive intervention significantly reduces long-term spread and minimizes future control costs (Epanchin-Niell & Hastings, 2021; Shackleton et al., 2022). Among the controls, treatment control  $u_4$  declines first after approximately 1 year, followed by animal movement control  $u_3$  after about 1.6 years, suggesting that these strategies are most effective when applied intensively during the initial phase of the control program when invasion pressure is highest.

The water management control  $u_2$  remains at full intensity for a longer period, up to approximately 4.3 years, before gradually decreasing, while land control  $u_1$  is maintained at maximum intensity for the longest duration, up to 6 years. This indicates that controlling mesquite spread through direct land management and water systems requires sustained intervention over a longer period compared to treatment and animal-related controls. Similar observations have been reported by Pyšek et al. (2023) and Zenni et al. (2022), who found that persistent habitat management and control of environmental dispersal pathways are essential for long-term suppression of invasive plant populations. The gradual decline of these controls reflects the reduced need for intervention as infestation levels decrease over time.

Overall, all control strategies continuously decline and reach zero intensity at year 10, demonstrating an optimal strategy where resources are heavily concentrated during the early years and progressively reduced as mesquite invasion is brought under control. In terms of relative importance, the duration of maximum effort follows the order land control  $u_1 >$  water control  $u_2 >$  animal control  $u_3 >$  treatment control  $u_4$ , suggesting that land and water management are the most critical long-term strategies, while animal and treatment controls are more effective as short-term early interventions. This pattern agrees with recent optimal control studies showing that adaptive time-dependent strategies improve ecological management efficiency by balancing intervention intensity with economic sustainability (Ahmed et al., 2024; Cuthbert et al., 2023).



### 5.5. Cost-Effectiveness Analysis

Cost-effectiveness analysis provides an important quantitative framework for evaluating ecological management strategies by simultaneously assessing economic expenditure and ecological benefit achieved under alternative intervention policies. In invasive species management, decision-makers are required to identify intervention strategies capable of minimizing invasion burden while maintaining economically sustainable implementation costs. In the present study, cost-effectiveness analysis was performed using the Incremental Cost-Effectiveness Ratio (ICER) framework to compare the economic efficiency of alternative mesquite management strategies. The economic costs considered in this analysis are expressed in United States Dollars (USD) to reflect realistic field implementation expenditures associated with mesquite management activities observed in Kahe Ward, where active invasion management interventions have been implemented under the CABI Prosopis management project. Similar cost-effectiveness approaches have been widely applied in ecological management and optimal control studies to evaluate long-term sustainability of intervention strategies (Lenhart and Workman, 2007; Asamoah et al., 2021).

#### 5.5.1 Incremental Cost-Effectiveness Ratio (ICER) Framework

The ICER framework is widely used in ecological economics and mathematical biology because it quantifies the trade-off between economic cost and ecological benefit when comparing competing intervention strategies. In invasive species systems, ICER measures the additional economic cost required to achieve a unit reduction in ecological invasion burden when shifting from one control strategy to another. This framework provides a useful decision-making tool for identifying economically efficient management policies under resource-limited environmental systems (Croft and Massei, 2022; Zahra et al., 2025).

#### 5.5.2 Total Control Cost Function

The total implementation cost associated with applying control interventions over the management period is represented by the cost functional

$$C(u) = \int_0^T \sum_{i=1}^4 \frac{1}{2} C_i u_i^2(t) dt$$



where  $u_i(t)$  denotes the intensity of control effort applied at time  $t$ , corresponding to land clearing ( $u_1$ ), water system management ( $u_2$ ), animal movement restriction ( $u_3$ ), and environmental treatment ( $u_4$ ), while  $C_i$  represents the cost coefficient associated with each intervention strategy. The total cost  $C(u)$  is measured in United States Dollars (USD). The quadratic cost formulation reflects increasing marginal implementation cost as intervention intensity increases and is commonly used in optimal control theory because it guarantees convexity and mathematical tractability of the optimization problem (Lev Pontryagin et al., 1962; Lenhart and Workman, 2007).

### 5.5.3 Ecological Burden (Invasion Impact)

The ecological burden represents cumulative ecosystem degradation resulting from persistent mesquite invasion over the simulation period. It is quantified as

$$E(u) = \int_0^T (L(t) + W(t))dt$$

where  $L(t)$  and  $W(t)$  represent mesquite invasion density in terrestrial land systems and aquatic ecosystems respectively. The resulting ecological burden is expressed in Invaded Area-Years (ha·yr), representing cumulative invaded ecological area measured in hectares integrated over time in years, thereby capturing long-term ecosystem pressure caused by persistent invasion. This ecological metric is widely used in environmental management studies because it measures cumulative habitat degradation associated with invasive species persistence (Barros et al., 2022; Roques et al., 2022).

### 5.5.4 Incremental Cost-Effectiveness Ratio (ICER)

To compare two competing intervention strategies  $X$  and  $Y$ , the Incremental Cost-Effectiveness Ratio is defined as

$$ICER_{(X,Y)} = \frac{C_X - C_Y}{E_Y - E_X}$$

where  $C_X$  and  $C_Y$  denote total implementation cost measured in USD, while  $E_X$  and  $E_Y$  denote ecological burden measured in Invaded Area-Years (ha·yr). The ICER therefore measures the additional economic cost required to reduce one additional unit of invaded ecological area over time. This framework has been extensively used in ecological decision analysis for evaluating the economic efficiency of alternative intervention strategies (Asamoah et al., 2021; Croft and Massei, 2022).





### 5.5.5 Simulation-Based Estimation of Cost and Ecological Burden

The economic cost and ecological burden values used in this analysis were obtained directly from numerical simulations of the mesquite invasion optimal control model. The nonlinear dynamical system describing the temporal evolution of land infestation  $L(t)$ , water infestation  $W(t)$ , animal-mediated seed carriers  $A(t)$ , and water-borne seed carriers  $F(t)$  was solved numerically over the simulation horizon  $T$ , together with the optimal control system derived using Pontryagin's Maximum Principle. Five intervention scenarios were considered, namely no control, mechanical control, chemical control, biological control, and combined integrated control strategy.

The economic cost assumptions were calibrated to reflect practical field implementation costs observed during mesquite management activities in Kahe Ward, where field observations showed active intervention measures including mechanical uprooting of mesquite stands, herbicide application, controlled livestock grazing management, restoration of degraded grazing lands, continuous ecological monitoring, and community mobilization programs undertaken under the CABI Prosopis management initiative. These field-based observations provided a realistic economic basis for calibrating the model cost coefficients used in the cost-effectiveness analysis.

### 5.5.6 Estimation of Total Economic Cost

The total intervention cost associated with each strategy was computed using

$$C(u) = \int_0^T \sum_{i=1}^4 \frac{1}{2} C_i u_i^2(t) dt$$

where the control trajectories  $u_i(t)$  were obtained directly from simulation outputs. Numerical integration of time-series control data was performed to estimate cumulative management expenditure over the intervention period. The resulting cost values are expressed in USD, representing realistic financial investment required for field implementation.

### 5.5.6 Estimation of Ecological Burden

The ecological burden associated with each intervention strategy was computed using

$$E(u) = \int_0^T (L(t) + W(t)) dt$$



where  $L(t)$  and  $W(t)$  were obtained from simulated invasion trajectories. Numerical integration over time produced cumulative ecological burden measured in Invaded Area-Years (ha·yr), representing total invaded ecosystem area maintained over the management horizon.

### 5.5.8 Comparative Cost-Effectiveness Analysis

Using the simulation-derived economic cost and ecological burden values, comparative cost-effectiveness analysis was performed to evaluate the relative economic efficiency of alternative mesquite management strategies. The intervention costs were calibrated to reflect observed field expenditures associated with invasive mesquite management operations in Kahe Ward, where field implementation under the CABI project involved substantial operational expenditure related to labor-intensive clearing operations, chemical herbicide procurement, biological suppression measures, transport logistics, ecological monitoring, and community awareness programs. The simulation-derived strategy comparison is summarized in Table 5.

**Table 5.8: Comparative Economic Cost and Ecological Burden Under Alternative Management Strategies**

| Strategy                             | Total Cost (USD) | Ecological Burden (ha·yr <sup>-1</sup> ) |
|--------------------------------------|------------------|------------------------------------------|
| No Control                           | 195.67           | 4.8921                                   |
| Mechanical Control                   | 320.15           | 3.1254                                   |
| Chemical Control                     | 262.83           | 3.4567                                   |
| Biological Control                   | 285.42           | 2.8341                                   |
| Combined Integrated Control Strategy | 347.54           | 2.1048                                   |

The results demonstrate that all intervention strategies substantially reduce ecological burden relative to the uncontrolled invasion scenario. Mechanical control significantly reduces mesquite spread but incurs relatively high operational costs because field implementation requires extensive labor, cutting equipment, uprooting operations, and biomass disposal. Chemical control provides moderate invasion suppression at relatively lower cost, whereas biological control achieves stronger ecological benefit through long-term suppression of mesquite regeneration. The combined integrated control strategy, representing simultaneous application of mechanical removal, chemical treatment, biological suppression, water pathway management, and continuous ecological monitoring, produced the lowest ecological burden of 2.1048 ha·yr, indicating



the greatest long-term reduction in cumulative ecosystem degradation. Although this strategy generated the highest implementation cost (USD 347.54), this result closely reflects field observations in Kahe Ward where integrated mesquite management interventions implemented under the CABI project required substantial financial investment due to equipment deployment, herbicide procurement, restoration activities, monitoring programs, and local community participation.

## 5.6 ICER Results and Economic Interpretation

The ICER calculations were performed to compare the economic efficiency of alternative intervention strategies.

Mechanical Control versus No Control

$$ICER = \frac{320.15 - 195.67}{4.8921 - 3.1254} = 70.46$$

Chemical Control versus No Control

$$ICER = \frac{262.83 - 195.67}{4.8921 - 3.4567} = 46.79$$

Biological Control versus No Control

$$ICER = \frac{285.42 - 195.67}{4.8921 - 2.8341} = 43.61$$

Combined Strategy versus Biological Control

$$ICER = \frac{347.54 - 285.42}{2.8341 - 2.1048} = 85.18$$

The results indicate that although the combined integrated strategy requires the highest economic investment, it produces the greatest ecological benefit by achieving the lowest cumulative invasion burden. Biological control demonstrates relatively strong economic efficiency, while mechanical control incurs higher operational expenditure despite significant invasion reduction. From a long-term ecological management perspective, the integrated strategy offers the most sustainable solution because it simultaneously targets multiple ecological pathways responsible for mesquite spread.



These findings are consistent with field observations in Kahe Ward, where mesquite invasion management has shown that reliance on isolated intervention strategies often leads to reinvasion, whereas integrated management approaches combining physical removal, chemical suppression, ecological restoration, and continuous monitoring provide more sustainable long-term ecological outcomes. Similar conclusions have been reported in recent ecological management studies demonstrating that adaptive integrated control strategies consistently outperform single intervention approaches when balancing ecological suppression against economic sustainability (Hu et al., 2024; Chhetri and Rathish Kumar, 2025; Zahra et al., 2025).

Overall, the results confirm that integrated adaptive control strategies provide the most cost-effective long-term approach for sustainable mesquite invasion management across coupled land–water ecosystems.

## 6. Discussion

The findings of this study demonstrate that integrated management strategies are more effective than single-method interventions in controlling mesquite invasion because invasion dynamics are sustained by multiple interacting ecological pathways operating simultaneously across terrestrial and aquatic ecosystems. Numerical simulations showed that although mechanical, chemical, and biological controls individually reduced invasion burden, the combined integrated strategy achieved the lowest ecological burden (2.1048 ha·yr), confirming that effective invasive species management requires simultaneous disruption of interconnected dispersal mechanisms. Similar studies have shown that invasive species management becomes significantly more successful when multiple intervention approaches are combined rather than relying on isolated control methods (Croft & Massei, 2022; Barros et al., 2022).

The optimal control analysis further revealed that time-dependent adaptive interventions provide superior ecological and economic outcomes compared with static control strategies. The computed control trajectories indicated strong early intervention followed by gradual reduction as invasion pressure declined, suggesting that dynamically adjusting control intensity improves long-term management efficiency while minimizing unnecessary operational cost. This finding agrees with recent dynamic optimal control studies which show that adaptive intervention strategies consistently outperform constant controls in complex ecological and invasion systems (Rui et al., 2022).



Sensitivity analysis identified land-based ecological pathways as dominant drivers of mesquite invasion persistence, particularly parameters associated with animal-mediated seed deposition ( $\eta_1$ ), runoff-driven transport ( $\alpha_2$ ), and flood-mediated dispersal ( $\rho_1$ ). This indicates that terrestrial interventions such as early detection, mechanical clearing, and monitoring of invasion corridors remain critical for long-term suppression. Similar landscape-scale ecological studies have demonstrated that land-use structure, disturbed habitats, and transportation corridors significantly accelerate invasive species spread, reinforcing the importance of land-based intervention strategies (Padilla et al., 2022; Roques et al., 2022).

The cost-effectiveness analysis showed that although the combined strategy incurred the highest implementation cost (USD 347.54), it generated the greatest ecological benefit and produced the most sustainable long-term management outcome. This finding is consistent with field observations conducted in Kahe Ward under interventions implemented through the CABI project, where isolated interventions frequently resulted in reinvasion, whereas integrated management approaches provided more durable suppression despite higher initial cost. Overall, these findings support recent invasion economics literature emphasizing that effective invasive species management requires adaptive, economically optimized, and integrated intervention strategies capable of addressing ecological complexity across multiple environmental pathways (Epanchin-Niell et al., 2022; Perrings et al., 2023).

## **7. Conclusion and Recommendations**

This section presents the overall conclusions and recommendations derived from the findings of the study on mesquite invasion dynamics and control strategies. It summarizes the key outcomes obtained from model formulation, theoretical analysis, numerical simulations, and optimal control evaluation, highlighting the major insights on invasion spread mechanisms and the effectiveness of intervention strategies. Based on these findings, practical recommendations are proposed to support sustainable management of invasive mesquite species and guide future research and environmental policy decisions.

### **7.1 Conclusion**

This study developed a mathematical model describing mesquite invasion dynamics across interconnected land and water ecosystems by incorporating terrestrial spread, hydrological transport, animal-mediated seed dispersal, and water-borne transmission pathways. Sensitivity analysis identified animal-mediated seed deposition, runoff-driven transport, and hydrological



pathways as dominant drivers of invasion persistence. Numerical simulations showed that mesquite invasion increases progressively in the absence of intervention, while optimal control analysis demonstrated that time-dependent adaptive interventions outperform static control strategies in reducing long-term invasion burden. Cost-effectiveness analysis further revealed that although the combined integrated strategy incurred the highest implementation cost USD 347.54 hectares per year ), it achieved the lowest ecological burden 2.1048 hectares per year, making it the most sustainable long-term management option. These findings, supported by field observations in Kahe Ward under interventions implemented through the CABI project, confirm that integrated adaptive control provides the most effective strategy for long-term mesquite management.

## 7.2 Recommendations

Based on the findings, this study recommends prioritizing integrated management strategies combining mechanical, chemical, biological, and monitoring interventions, since combined control produced the greatest ecological benefit. Management efforts should focus on land-based invasion pathways, animal movement corridors, and seasonal water channels, which were identified as critical drivers of invasion spread. In addition, environmental management programs should adopt adaptive time-dependent intervention strategies rather than fixed control approaches to improve long-term ecological and economic efficiency. Future research should incorporate spatial dynamics, climate variability, and field-based parameter estimation to improve model realism and strengthen large-scale invasive species management planning.

## Declaration

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**Data Availability:** All data are contained within the article

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